

Samuel Xavier Borges

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EDUCATION

University of Central Florida

2024 – 2028

Bachelor of Science in Computer Science, Minor in Data Science, Minor in Studio Art | GPA: 3.5

Orlando, FL

EXPERIENCE

Software Engineering Intern

Feb 2025 – Jan 2026

Knight Hacks

Orlando, FL

- Improved site accessibility and maintainability by developing reusable **React/TypeScript** UI components to replace non-semantic SVG elements and embed proper **ARIA** attributes, enhancing keyboard navigation compliance and cutting UI code duplication.
- Developed a ground-up email automation system using the **Google API** to send personalized acceptance, denial, and event-detail communications to applicants for one of **Florida's largest hackathons**.

Software Engineering Intern

May 2025 – Aug 2025

Data-Enabled Photovoltaics

Orlando, FL

- Automated parsing and normalization of heterogeneous instrument metadata by developing an **object-oriented Python** parser with **Pandas** and **SQLite**, reducing manual preprocessing time by over **75%**.
- Achieved consistent metadata provenance and context tracking by defining and validating **FAIR** data model standards and integrating workflow outputs across pipelines.

Software Engineering Intern

Mar 2025 – May 2025

Zuleris Interactive

Orlando, FL

- Developed a modular **Unity/C# radio interference simulator** with configurable overlap frequencies and **coroutine-driven** randomized jamming, enabling a fully adjustable multi-radio training tool.
- Implemented global static **C#** events to broadcast **UI** open/close and frequency changes, synchronizing audio playback and tuning across **four** independent radio instances for seamless, consistent behavior.

Software Engineering Intern

Jun 2023 – Aug 2023

Miami EdTech

Miami, FL

- Assisted with the testing and development of **visual recognition AI** software in **Python**, increasing efficiency & accuracy by **25%**.
- Researched the possible integrations & benefits of STEM education curriculum packages as part of a team of interns, suggesting a proposal to reach **30+** schools in the surrounding area.

PROJECTS

Ctrl-ARM | *Python, Pandas, Electron, React, Google ADK, TensorFlow, Whisper*

- Created a body movement **accessibility** system that combined **EMG** signals with **IMU** motion data and a multimodal **Electron + React** overlay, achieving **1st at ShellHacks 2025's Microsoft AI Challenge**.
- Engineered **Pandas ETL** pipelines to ingest and transform 200Hz EMG/IMU streams, enabling per-user calibration, feature extraction, and consistent **90%** classification accuracy across **12+** gestures.

Tariffix | *TypeScript, React, Vite, Tailwind CSS, MongoDB, Chrome Extension API, Gemini API, Cheerio*

- Enabled real-time tariff insights on **100K+** product listings via a **Vite-powered Chrome extension** backed by an **AI scraping pipeline** for metadata and HTS codes, winning **Bitcamp 2025's "Best use of MongoDB."**
- Delivered tariff cost estimates against official U.S. rates with under **5%** average error by comparing **AI-scraped data** to a **Gemini-powered MongoDB dataset**, automating precise calculations for end users.

Kmodo | *TypeScript, Next.js, React, tRPC, Tailwind CSS, Drizzle, PostgreSQL*

- Engineered full-stack web applications by integrating responsive user interfaces with scalable backend services, leveraging modern frameworks like **Next.js**, **tRPC**, and **PostgreSQL**.
- Self-hosted infrastructure on a **Linux VPS** with a CI/CD script for reliable deployment and extensibility.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, HTML, CSS, JavaScript, TypeScript

Libraries & Frameworks: Pygame, React, Next.js, Node.js, Vite, Tailwind CSS, tRPC, PostgreSQL, MongoDB

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Eclipse, Unity, Vercel, Prisma, Drizzle, Figma